

Applications of Statistical Modeling:
Module #1, Lecture #4

Alternative Statistical Models for Longitudinal Data:
A Conceptual Overview

Instructor: Brady T. West
Email: bwest@umich.edu

Lecture Overview

- Definitions of Longitudinal and Repeated Measures Data, and Data Structure
- Hierarchical vs. Marginal Approaches
 - Multilevel Modeling
 - Marginal Linear Models for Normal Outcomes
 - Generalized Estimating Equations
- Non-Normal Outcomes
- **Supplement**: See software notes on Canvas
- Take-Home Points

Repeated Measures / Longitudinal Data

- **Repeated measures** in general refer to repeated measures of the same outcome collected from the same units of analysis
 - Could be repeated measures under different experimental conditions / treatments
 - Could be repeated measures over time
- **Longitudinal data** represent a *special case* of repeated measures data: repeated measures over **long periods of time** on the same units
 - Our focus in this portion of the class (panels)

Repeated Measures / Longitudinal Data

- The key point in thinking about statistical models for longitudinal data is that we need to estimate parameters that capture **correlations of the repeated measures**
- This can be done by fitting models with random effects, correlated errors, or both!
- We will talk about these different approaches today and next week

Traditional Approaches

- Traditional / classical approaches to the analysis of repeated measures data rely on **repeated measures analysis of variance (rmANOVA)**
- This is a **multivariate** analysis approach, where the repeated measures arise from a **balanced design** (same time points / conditions for all subjects, no missing data)

Traditional Approaches

- rmANOVA approaches have many **limitations**:
 - A wide data structure is needed, with complete data for all subjects in the variables containing the repeated measures
 - Generally relies on only one type of marginal covariance structure for the repeated measures (“spherical”, or compound symmetry)
 - Listwise deletion means that subjects with any missing data will be dropped from the analysis!
 - Non-normal outcomes are not easily handled
 - Time-varying covariates are not allowed!

“Modern” Approaches

- More flexible “modern” approaches have been enabled by rapid developments in statistical software over the past few decades
- These approaches can accommodate data that are missing at random, and enable analysts to consider a variety of covariance structures
- We will discuss three general classes of models, but we first need to consider the appropriate data structure for longitudinal modeling

Appropriate Data Structure

- In general, statistical software for fitting longitudinal models requires a vertical data structure:

ID	Time	Group	DV	TVwgt	BASEwgt
1	1	1	180	15	10
1	2	1	185	18	10
1	3	1	195	22	10
...	...	...	...	...	...
2	1	0	176	23	20
2	2	0	167	25	20
2	3	0	168	31	20

Appropriate Data Structure

- Note the “vertical” or “long” structure, with multiple rows per subject / unit ID
- Sometimes referred to as a “unit-time” or “subject-time” data structure
- Contrast this with the “wide” structure, with one row per person and multiple variables containing the repeated measures
- This “univariate” structure takes full advantage of the available information

Appropriate Data Structure

- Note also the presence of a **baseline sampling weight**, representing a population weight at the time that a person entered a sample
- We also see a **time-varying weight**, representing an adjusted version of the baseline weight accounting for differential attrition at individual time points
- Original stratum and cluster variables would also be included in the data set (not shown in the previous data table)

Introducing the HRS Data

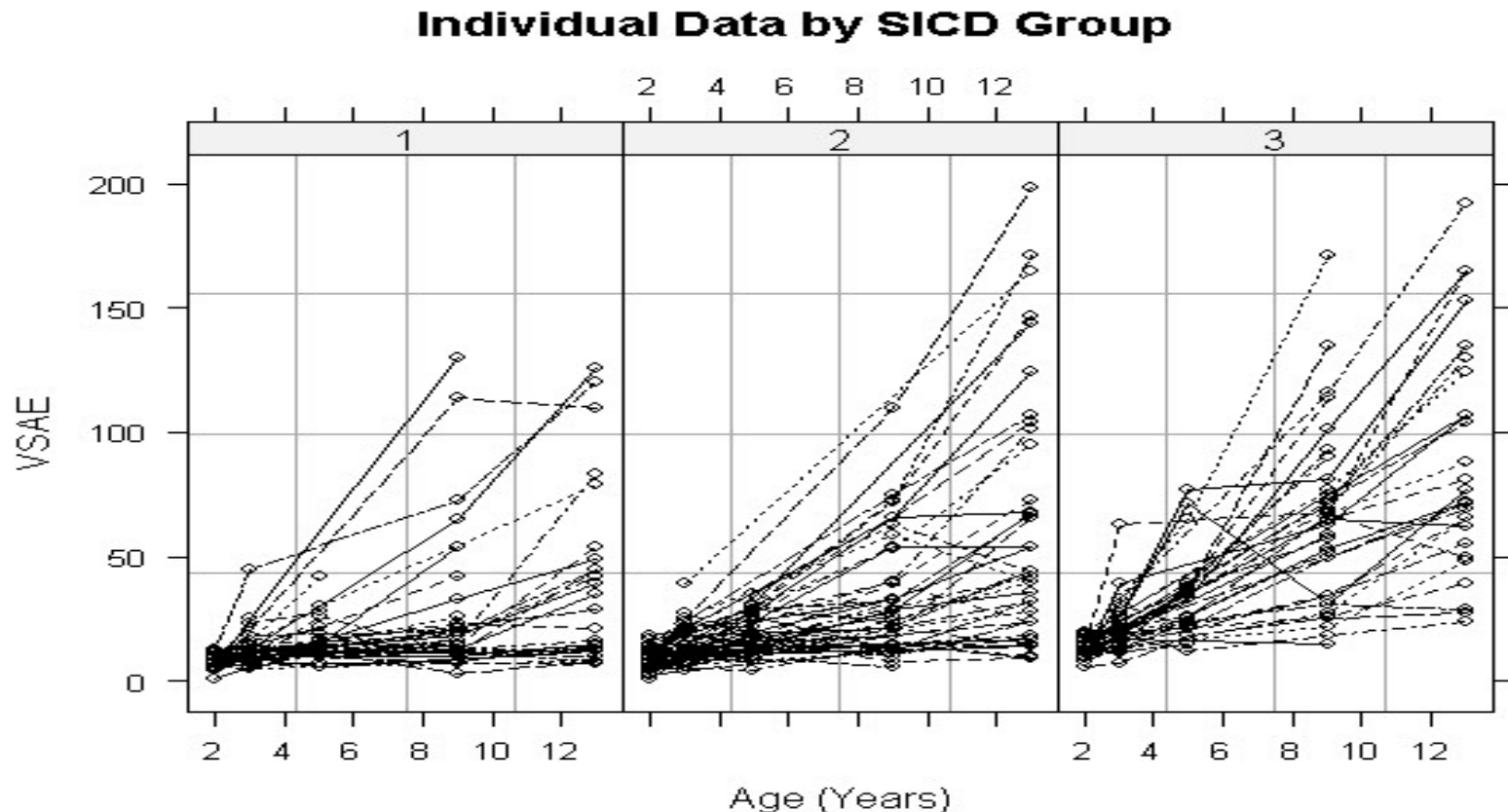
- On Canvas, you'll find R, SAS, Stata, and Excel versions of a file containing data from four waves of the Health and Retirement Study (HRS); the file names are hrs_reduced2.xxx
- Waves: 2000, 2002, 2004, 2006
- Note the “long” format of the data file
- We see data on total household assets, and demographics for the head of the household
- We also note complex sampling features!

First Things First...

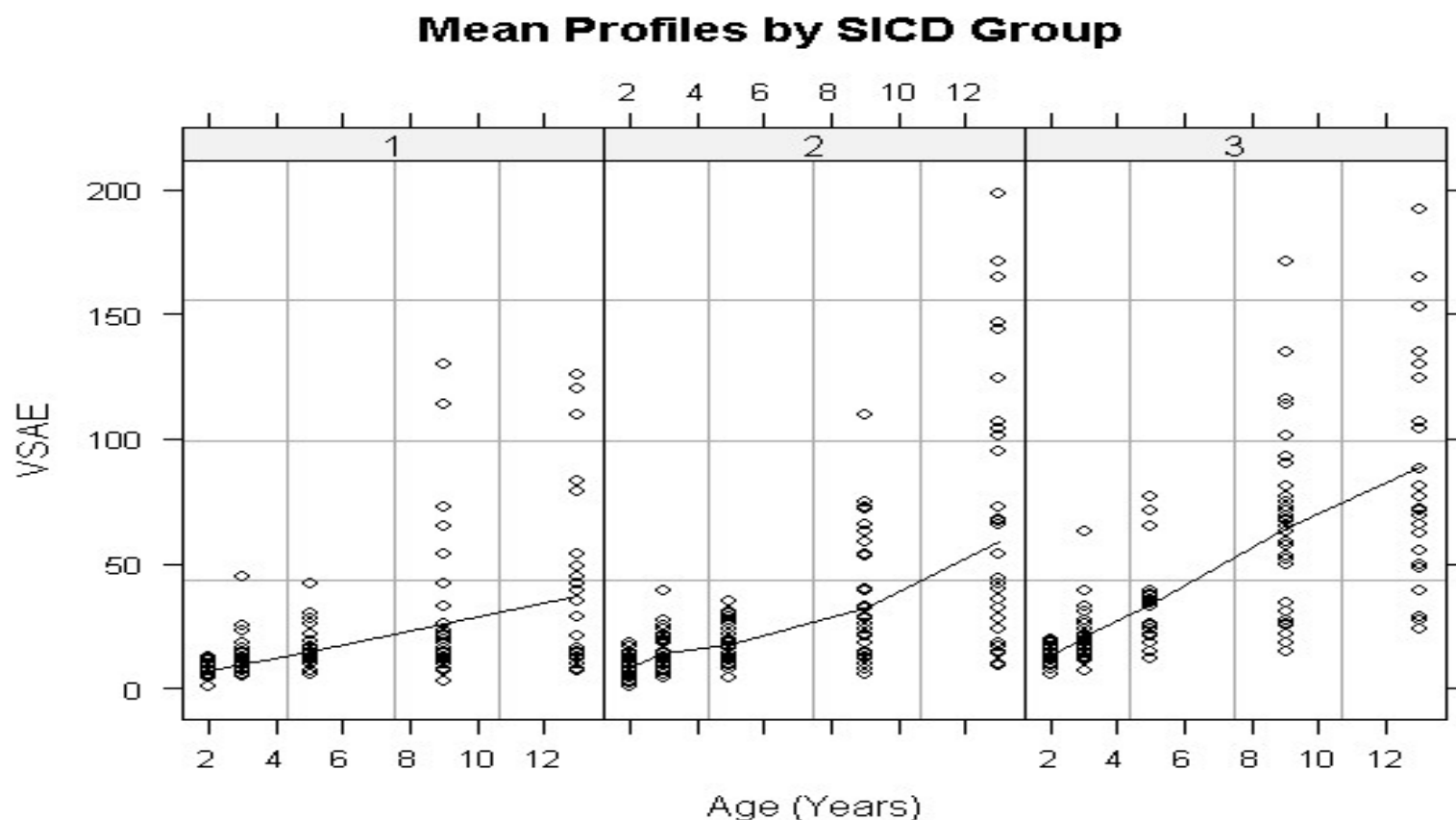
- Regardless of the modeling approach that you decide to use when analyzing longitudinal data, **always start with descriptive plots**
- “Spaghetti” plots are generally best, showing variation in trends among units based on the available information (or variation in means among subgroups); see the `ggplot` code from last week!
- These plots are easy to generate in various general-purpose software packages (see today’s **software supplement** on Canvas)

Example: Plotting the Data

- From West et al. (2022, *Linear Mixed Models*):



Example: Plotting the Data



Alternative Modeling Approaches

- We will talk about the following classes of models over the next two weeks:

Outcome	Multilevel	Marginal	
Normal	HLM	GEE	Marginal LM
Non-Normal	HGLM	GEE	

- Multilevel and marginal models are mainly distinguished by the presence or absence of **random effects**, which account for between-subject variance in coefficients of interest

Multilevel Modeling Approaches

- Recall the general specification of a multilevel model for a normal outcome, with t = time point, and i = subject:

$$y_{ti} = \beta_0 + \beta_1 x_{1ti} + u_{0i} + u_{1i} x_{1ti} + e_{ti}$$

$$\begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_0^2 & \sigma_{01} \\ \sigma_{01} & \sigma_1^2 \end{pmatrix} \equiv D \right)$$

$$e_{ti} \sim N(0, \sigma^2)$$

Multilevel Modeling Approaches

- Also recall the **matrix specification** of this model, both for the **conditional model** and the **implied marginal model** (what we fit!):

$$y_i | u_i \sim N(X_i\beta + Z_i u_i, R_i)$$

$$y_i \sim N(X_i\beta, Z_i D Z_i' + R_i \equiv V_i)$$

- When analyzing **longitudinal data**, we can consider more flexible structures for R_i , given that conditional errors may be correlated!

Multilevel Modeling Approaches

- The key here is that by using **explicit random effects**, we approximate subject-specific longitudinal profiles with unique linear regression functions
- The multilevel model thus has two levels:
 - **Level 1:** A model for each subject, relating time-varying predictor variables to the dependent variable
 - **Level 2:** A model for each subject-specific coefficient, explaining variance in the **random coefficients** with subject-level predictors

Multilevel Modeling Approaches

- Note that the fixed effects in these models **only appear at Level 2** of the model specification!

Level 1: $y_{ti} = \beta_{0i} + \beta_{1i}x_{1ti} + e_{ti}$

Random coefficients (not parameters!)

Level 2: $\beta_{0i} = \beta_0 + u_{0i}$

$$\beta_{1i} = \beta_1 + u_{1i}$$

We add subject-specific covariates and their fixed effects to the Level 2 equations to explain variance in the random effects!

Multilevel Modeling Approaches

- For example, suppose that we add fixed effects of a specialized treatment (a binary, time-invariant covariate), randomly assigned in a longitudinal clinical trial, to the Level 2 equations:

Level 1: $y_{ti} = \beta_{0i} + \beta_{1i}x_{1ti} + e_{ti}$

Level 2: $\beta_{0i} = \beta_{00} + \beta_{01}T_i + u_{0i}$

$$\beta_{1i} = \beta_{10} + \beta_{11}T_i + u_{1i}$$

Note that we would expect the variances of the random effects to be reduced. But how do we interpret the 4 parameters?

Multilevel Modeling Approaches

- As mentioned earlier, with longitudinal data, we can consider more general structures for the R_i matrix, which captures the variances and covariances of conditional errors for the same subject i
- **Some Examples:**
 - **First-order auto-regressive** (for equally-spaced time points; constant variance, correlation that weakens as a power function of the time between different points)
 - **Simple diagonal/spherical** (constant variance, zero covariance)
 - **Compound symmetry** (constant variance, constant covariance)
 - **Heterogeneous variance** (error variance changes as a function of the time point, or as a function of different subgroups)

Multilevel Modeling Approaches

- Later, we'll consider how multilevel models might change when outcomes are not normally distributed
- Now, we consider alternative approaches for fitting the **marginal model**, where random effects are not explicitly included (different parameterization of the V_i matrix!)

Marginal Modeling Approaches

- A researcher may not care about estimating between-subject variability in slopes or intercepts; only fixed effects are of interest!
- They may only be interested in overall marginal trends, possibly for different groups
- Alternative methods exist for fitting these types of **marginal** models:
 - Marginal Linear Models (MLM)
 - Generalized Estimating Equations (GEE)

Marginal Modeling Approaches

- When fitting marginal models to normal dependent variables, we estimate parameters defined by the following model (no random effects, just errors):

$$y_{ti} = \beta_0 + \beta_1 x_{1ti} + \dots + e_{ti}$$

$$y_i \sim N(X_i \beta, V_i)$$

- Note that the variance-covariance matrix V_i is completely general, and can take on many possible structures (e.g., unstructured, AR(1), etc.)
- **But we do not explicitly estimate the variances and covariances of random effects in the D matrix!**

Marginal Linear Models

- With normal outcomes, we can estimate marginal models using a two-step approach (same idea used for multilevel models):
 - Estimate the parameters defining the variance-covariance matrix V_i using REML or ML
 - Use **Generalized Least Squares** to compute estimates of the fixed-effect parameters:

$$\hat{\beta} = \left(\sum_i X_i' \hat{V}_i^{-1} X_i \right)^{-1} \sum_i X_i' \hat{V}_i^{-1} y_i$$

Marginal Linear Models

- **Important:** If you fit a marginal linear model to a normal outcome ***using a multilevel (or mixed) modeling software procedure***, you are not actually fitting a mixed (or multilevel) model (recall the special case only when they coincide)!
- **Why?** You aren't including explicit random effects in the model; you are just allowing errors to correlate
- It is therefore important to refer to these models as *marginal linear models* (with correlated errors) in publications and reports, so that readers are not mislead about the methods used!

Generalized Estimating Equations

- When fitting models to longitudinal data using GEE, we seek estimates of parameters that solve the following **score function** (or **estimating equation**):

$$S(\beta) = \sum_{i=1}^n D_i^T V_i^{-1} (y_i - \mu_i) = 0$$

- D_i^T is an $n_i \times p$ matrix with (i,j) th elements $\frac{\partial \mu_{ti}}{\partial \beta}$
- V_i is the $n_i \times n_i$ variance-covariance matrix for the observations on subject i (user-specified)

Generalized Estimating Equations

- y_i is the vector of outcome measures collected for the i -th subject
- μ_i is the vector of expected means for the outcome measures **based on the specified model** (here is where the betas enter...)
- These equations are generally solved using either:
 - Iteratively weighted least squares, or
 - Fisher scoring algorithms (more common)

Generalized Estimating Equations

- The GEE methodology was first introduced by Liang and Zeger in a seminal 1986 paper (see Canvas)
- We seldom know the true variance-covariance matrix for the observations, so in practice we work with a **working correlation matrix** (a “plausible guess” for the true structure)
- We typically specify correlations and not covariances, because for non-normal outcomes the variances are defined by the mean structure of the model
- **This allows the methodology to be easily adapted to non-normal outcomes** (Poisson, Binomial, etc.)

Generalized Estimating Equations

- Variances of the parameter estimates are computed using what is known as a **sandwich estimator** (Liang and Zeger, 1986)
- This estimator is based on the specified working correlation matrix (e.g., exchangeable), and the variance-covariance matrix of the observations based on expected means from the fitted model
- Estimators of the beta parameters are **consistent** even if the specified working correlation matrix is incorrect; bad choices affect the standard errors

Generalized Estimating Equations

- Possible choices for the working correlation matrix in models fitted using GEE:
 - Independence (0 correlation, independent obs.)
 - Exchangeable (constant correlation)
 - AR(1) (first-order auto-regressive, decaying correlation over time)
 - Unstructured (completely general correlations)
- Estimators of the correlations were proposed by Liang and Zeger (1986)
- We generally focus on the betas, and do not make inferences about the “nuisance” correlations

Generalized Estimating Equations

- Inference that is “robust” to possible variance-covariance matrix misspecification is possible when using **Wald Tests** based on the sandwich estimates of the standard errors
- An adapted information criterion (QIC) can be used to compare the fits of competing models
- Good specification of the mean structure (interactions, non-linear terms, etc.) is important!
- With large data sets, alternative choices of the working correlation matrix do not make a large difference, but should still be considered

Generalized Estimating Equations

- Because poor choices of the working correlation matrix can affect standard errors (and thus inferences), it is worthwhile to try and choose a reasonable model based on information criteria
- The QIC, while popular and widely implemented, has been demonstrated to have important shortcomings in choosing the “correct” correlation structure
- Westgate and Burchett (2017; see Canvas) describe better alternatives for identifying the correct correlation structure, and provide R code to both fit GEE models and compute the advocated information criteria

Models for Non-Normal Outcomes

- In many longitudinal studies, outcome variables are not normally distributed
 - Binary indicators (drug use in past 3 months?)
 - Counts (frequency of drug use in past 3 months?)
 - Categorical outcomes (household income bracket)
 - Others?
- We therefore need to consider adaptations of the methods discussed so far for these types of outcome variables

Models for Non-Normal Outcomes

- GEE methods were specifically designed to readily accommodate non-normal outcome variables measured longitudinally
- The mean structure in the estimating equations is defined by a given type of generalized linear model
- **Example:** logistic regression for a binary DV

$$\mu_i = \pi_i = E(y_i | X_i) = \frac{\exp(X_i\beta)}{1 + \exp(X_i\beta)}$$

Models for Non-Normal Outcomes

- When considering multilevel models for non-normal outcomes (e.g., multilevel logistic regression), which include **explicit random effects**, the estimation methods are not as straightforward
- We will first describe the model, and then discuss alternative estimation procedures that are used in the software

Models for Non-Normal Outcomes

- We assume that given a vector of random effects for subject i , $\mathbf{u}_i$, all responses y_{ti} on the dependent variable are independent, and follow a density function that belongs to the **exponential family**:

$$f(y_{ti} | \theta_{ti}, \phi) = \exp \{ \phi^{-1} [y_{ti} \theta_{ti} - \psi(\theta_{ti})] + c(y_{ti}, \phi) \}$$

- The regression parameters and random effects enter as follows:

$$\theta_{ti} = \mathbf{x}'_{ti} \boldsymbol{\beta} + \mathbf{z}'_{ti} \mathbf{u}_i, \text{ with } \mathbf{u}_i \sim N(0, D)$$

Models for Non-Normal Outcomes

- The marginal distribution of the vector of responses on subject i is then found by integrating the product of the individual density contributions with respect to the random effects:

$$f_i(y_i | \beta, D, \phi) = \int \prod_{t=1}^{n_i} f(y_{ti} | u_i, \beta, \phi) f(u_i | D) du_i$$

- Given this marginal distribution, we can now write the likelihood function...

Models for Non-Normal Outcomes

- Likelihood function (in theory):

$$L(\beta, D, \phi) = \prod_{i=1}^n f_i(y_i | \beta, D, \phi)$$

- With a normal dependent variable, we can work out the necessary integral analytically
- With a non-normal dependent variable, good luck...=)
- We need to first **approximate** the likelihood!

Models for Non-Normal Outcomes

- We can approximate the likelihood using two possible approaches:
 - Approximation of the integrand
 - Approximation of the integral: also known as **numerical quadrature**
- The first approach basically requires approximating the inner product for a subject with a normal density; this requires a roughly continuous response, and many subjects!

Models for Non-Normal Outcomes

- More often in practice, we use **Gaussian Quadrature**, and replace the integral with a weighted sum of density functions
- The number of elements used in the sum is known as the number of **integration points**, where a higher number of points produces a more accurate approximation
- Algorithms are now available in software for determining these weighted sums

Models for Non-Normal Outcomes

- **Adaptive Gaussian Quadrature** is often used for this approximation, and needs fewer integration points than classical quadrature, but requires more computational time
- **Laplace Approximation** and **Penalized Quasi-Likelihood (PQL)** provide other options for approximating the integral (see Kim et al., 2013)
- Once we have approximated the likelihood, we can use standard maximization procedures

Models for Non-Normal Outcomes

- So which approximation method is best?
- Kim et al. (2013) simulations, on Canvas:
 - PQL estimates tend to be noticeably biased (this has also been found in other studies)
 - SAS PROC GLIMMIX – Laplace tended to perform well compared to other options
 - AGQ required substantial computational resources in most cases, but still works well
 - Use AGQ with smaller sample sizes or studies with fewer time points; Laplace superior with large samples

Take-away Points

- The difference between marginal and hierarchical (or multilevel) modeling approaches is extremely important when analyzing longitudinal data
- **Hierarchical:** include random effects, and inferences are subject-specific (i.e., you are “controlling” for random subject effects, and fixed effects represent what happens when a covariate changes for a given subject)
- **Marginal:** Population-averaged effects (what happens across all subjects in a study?)

Take-away Points

- For non-normal outcomes, GEE models are fairly straightforward to estimate, and do not involve the computational issues associated with random effects
- However, we are unable to make inferences about between-subject variability
- **Important question:** how should we model fixed time effects (Spline? Polynomial? Categorical?)? Stay tuned!

Thinking Ahead...

- Please review today's **software supplement**, and start fitting some initial models to the data set that you want to work with!
- Next week, we will consider a case study involving analysis of the HRS data
- In the final lecture of this module (after the Midterm Exam), we will discuss recent research on handling complex sample designs in models for longitudinal data